

Each question is worth 5 points for a total of 40 points. The Extra Credit problem can get you up to an additional 5 points. NO CALCULATORS allowed. If you use a calculator, the entire test will receive a score of 0.

Problem 1

Find the slope of the curve $f(x) = 8 - x^2$ at the point $(2, 4)$, and the equation of the tangent line through the given point.

Problem 2

Find

$$\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$$

Problem 3

Using the definition of a limit, prove that

$$\lim_{x \rightarrow 5} (x + 4) = 9$$

Problem 4

Find

$$\lim_{x \rightarrow -0.5^-} \sqrt{\frac{x+2}{x+1}}$$

Problem 5

Let $f(x) = \sin(x - \sin x)$. Find $\lim_{x \rightarrow \pi} f(x)$ and determine if the function f is continuous at π .

Problem 6

Let $f(x) = \frac{x+2}{x^2+5}$. Find $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$.

Problem 7

Find

$$\lim_{x \rightarrow -3} \frac{2 - \sqrt{x^2 - 5}}{x + 3}$$

Problem 8

Find $\lim_{x \rightarrow -2^+} \left(\frac{x}{x+1} \right) \left(\frac{2x+5}{x^2+x} \right)$

Extra Credit:

Prove that $\lim_{x \rightarrow c} f(x) = L$ if and only if $\lim_{h \rightarrow 0} f(c + h) = L$.